

THILAK D

Chief of Staff and Data Scientist

I work on a healthcare AI product at Mondee. I build and test the AI, and I also run the pilots, compliance work and planning around it.

Hyderabad, India. Open to roles anywhere in India, onsite, hybrid or remote.

PROFILE SUMMARY

I'm a data scientist working as Chief of Staff at Mondee in Hyderabad, on our healthcare AI product. I built the evaluation platform we run every release, analysed 50,000+ hospital encounters, fine-tuned a Telugu speech recognition model, and coordinate our pilots in six hospitals. I also work on the CDSCO filing and business planning with leadership.

Before Mondee, I spent a year and a half part-time at Sentinel AI ML Labs, much of it during my master's, building AI for healthcare and MedTech clients and mentoring interns. My first-author paper, SAGE, combines knowledge graphs and vector search for healthcare information retrieval, and I'm doing an MBA in Strategy and Consulting at BITS Pilani alongside work. I want a role where I keep building with data and AI while working close to leadership decisions.

CONTACT

+91 63695 02212

saithilak04@gmail.com

linkedin.com/in/thilak-d-129464205

github.com/SaiThilak042k3

thilakd.vercel.app

Hyderabad, Telangana

LANGUAGES

Tamil, native

English, fluent

Telugu, conversational

INTERESTS

Video editing, music production, reading tech blogs, badminton, cricket, volleyball

WORK EXPERIENCE

Chief of Staff (Data Science and AI)

Oct 2025 – Present

Mondee Tech Private Limited (Mondee Tabhi Group), Hyderabad

I report to the Group Chief Growth Officer and work across our healthcare AI product, on the data science and on the pilots, compliance and planning around it.

AI evaluation platform

- Built our evaluation platform with the data science team. It checks every AI summary against the doctor's edited version, scores ASR and LLM output, and runs each release so the product team can review AI and engineering work.

Healthcare AI: data science and product

- Work on the data science, product and rollout of our clinical documentation product, where doctors check and correct AI summaries field by field.
- Analysed 50,000+ patient encounters and 300+ doctor records and found that paused consultations were later marked finished with no record of whether they resumed. Worked with the dev team on the fix, and that error is now close to zero.
- Noticed many doctors were leaving paused cases unreviewed and changed the review flow: doctors now confirm each review, and refresh and back are blocked during a live session.
- Fine-tuned Google's MedASR on 15,000 Telugu consultation recordings. The vendor we already used was still more accurate (7.5% vs 10.5% word error rate, 2.8% vs 4.2% on medical terms) and faster, so we stayed with it. Pipelines for Hindi, Tamil and Bengali are ready.
- Built a React and FastAPI dashboard on doctor review data that all six pilot hospitals now use. It tracks KPIs and sends flagged fields to the AI, product or dev team to fix.
- Coordinate our pilots across six hospitals, including government hospitals. In one pilot, 4 doctors reviewed about 3,000 records over a month and accepted 88% of AI summaries without edits. Their main feedback was that sorting records took too long.

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WORK EXPERIENCE, CONTINUED

Chief of Staff (Data Science and AI), continued

Oct 2025 – Present

Mondee Tech Private Limited (Mondee Tabhi Group), Hyderabad

Healthcare AI: data science and product

- Planned a set of small in-house models, one per data science task, so we can build our own AI stack and personalise the product.
- Started a new Ayurveda AI project: a fine-tuned small language model with a RAG system, from data collection to the training pipeline.
- Lead a team of 6 that tests the patient and doctor apps (Android, iOS, web) before every release and follows up on fixes with the dev and AI teams. Also keep the release tracker leadership uses; we ran 3 release cycles over the past year.
- Mapped 30,000 doctors across Hyderabad and shortlisted 2,000 who also consult part time at their own clinics, as the pool for our care network.

Regulatory and digital health

- Prepared our CDSCO application for the product as a Class A medical device: post-market surveillance plan, PSUR, ISO 13485 internal audit and the MD-12 test licence, which is now under review.
- Handle our ABDM integration: briefed the team, built the prototype through milestones M1 to M4, and worked with legal on the patient consent form.

Strategy and growth

- Built the capex, opex and 3-year P&L model for a planned USD 100M care network: 4 inpatient hubs, 80 diagnostic centres, and mobile doctor and nurse units. The plan is now being implemented.
- Drafted the UAE market entry plan and senior outreach to government and investment bodies for the group's travel business.
- Drafted and managed an MoU with a liaison consultancy to open government and private sector work in North India (aviation, GCCs, urban development).
- Leading a workshop for a national nursing association in Delhi and Noida (upcoming).

WORK EXPERIENCE, CONTINUED

Sentinel AI ML Labs (SAIMA Labs)

Jun 2024 – Nov 2025

Part-time. Chennai. AI consulting for healthcare, MedTech and startups.

Lead, AI ML Special Projects (AI Data Science Analyst)

May 2025 – Nov 2025

Junior Data Scientist

Jun 2024 – May 2025

- Built a dental health prediction model and automated WhatsApp messages to patients for a dental hospital group.
- Studied patient flow, scheduling and IT at a charitable dispensary in Chennai and suggested changes to their workflow.
- Built a risk scoring bot (RAS) that rates startups for an investment foundation, using 20+ of its datasets. It works for one startup or a bulk CSV.
- Worked with cardiologists on an AI cardiac risk prediction model for a MedTech startup.
- Helped pitch and present a cardiac MedTech startup's artificial heart project at innovation conferences in Colombo and Manila.
- Built and deployed two websites from scratch.
- Mentored 5 interns across the RAS bot, product and website projects.
- Made branding videos and creatives for a client's product launch in Dubai, and pitch decks for international MedTech events.

Data Analyst (internship)

May 2024 – Jun 2024

TaskBench, Chennai

- Built prediction and classification models in Python for market research, improving efficiency by 50%.
- Analysed 50,000+ records to give clients input for their marketing plans.
- Worked on 2+ digital marketing projects.
- Prepared 2 due diligence reports and 3+ client proposals.
- Handled several projects at once and delivered all of them on time.

WORK EXPERIENCE, CONTINUED

Freelance developer and designer
Jun 2023 – Present

Independent, alongside work and study

- Building a networking app for a client in React Native (iOS and Android).
- Built websites for individuals and organisations, including a multilingual WordPress site.
- Designed resumes for job seekers, including people later hired at State Street and Hexaware.
- Edited videos and ran YouTube channels for non-profits and educational groups.

VOLUNTEER EXPERIENCE

Web development and media volunteer
May 2021 – Mar 2022

Sri Sathya Sai Seva Organisation, Tamil Nadu

- Added multi-language support to the organisation's official WordPress website.
- Edited 20+ videos for the Sri Sathya Sai Balvikas children's YouTube channel.

LEADERSHIP AT SSSIHL

Student President, Digital Enlightenment Club — Jul 2023 – May 2025
Nandigiri Campus. Ran tech awareness and digital literacy work for students.

Co-founder, Eco-Einsteins Club — Feb 2024 – Apr 2025
Environmental and sustainability work on campus.

Member, Institute Innovation Council — 2024

Captain, Annual Sports and Cultural Meet — 2025

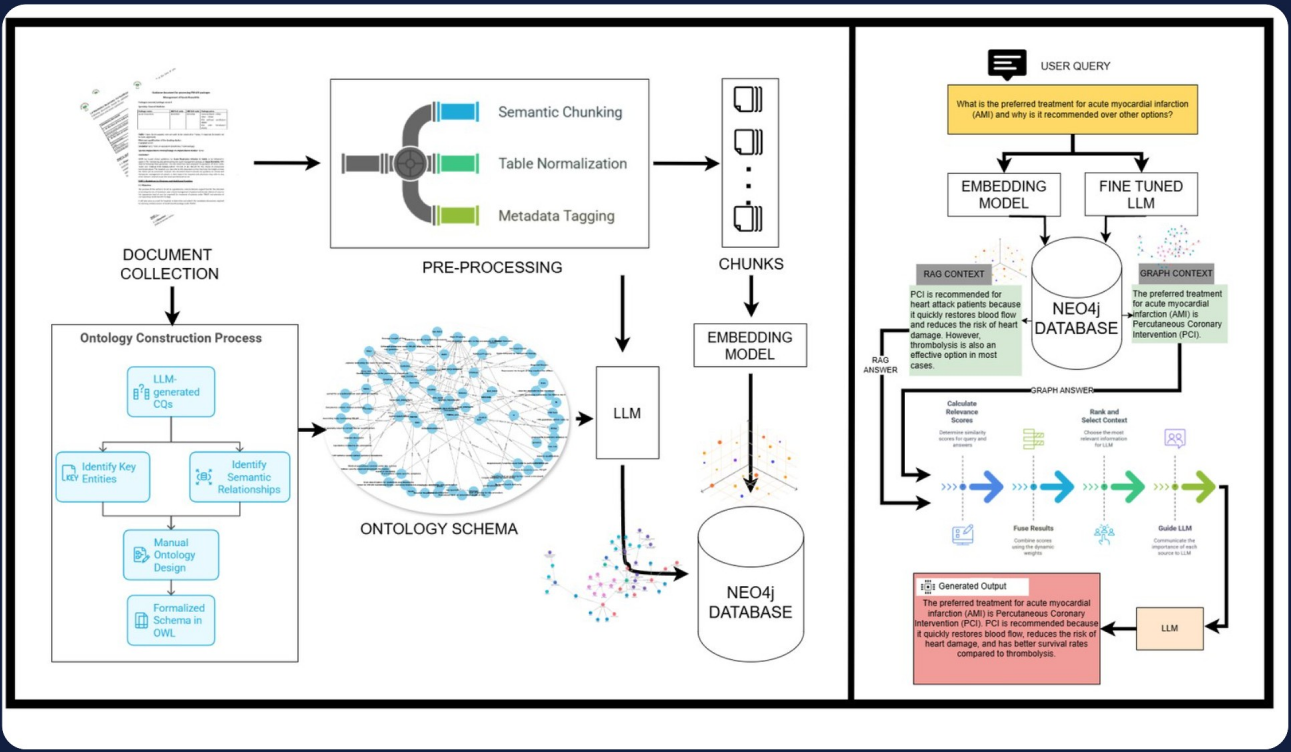
RESEARCH

Conference paper, first author

SAGE (Semantic and Graph-based Engine): hybrid knowledge retrieval for Ayushman Bharat treatment guidelines

Thilak D, P Sunil Kumar, SVSN Sarma. Presented at the 13th International Conference on Contemporary Engineering and Technology (ICCET 2025), Chennai, 22–23 March 2025. Paper ID ICCET250837.

Figure from the paper: ontology and preprocessing (left) feeding hybrid graph and vector retrieval (right)



- Fine-tuned LLaMA 3.1 (8B) and Phi-4 to turn plain questions into graph queries (Cypher). Phi-4 did better, with 0.86 BLEU and 0.73 exact match.
- Combined graph search with vector search. Answer relevance went from 0.89 with plain RAG to 0.95, and faithfulness from 0.87 to 0.94.
- Designed the ontology in OWL, had it checked by domain experts, and used an LLM to fill the Neo4j graph from it.

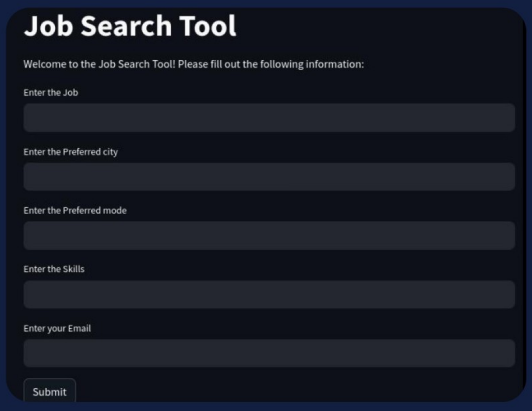
PROJECTS

Hybrid question answering with LangGraph: multi-hop RAG and knowledge graphs

MSc thesis, SSSIHL. Jun 2024 – Apr 2025. Guided by Prof. SVSN Sarma and Prof. P Sunil Kumar

- Extended the paper into the MSc thesis: a LangGraph agent that answers Ayushman Bharat questions, like which private hospitals in a state do a given surgery and what documents to bring.
- Collected 150+ scheme documents using RPA and built a knowledge graph of 50,000+ hospitals, 1,600+ surgeries and 9 states.
- The agent decides for each question whether to use the graph, the documents or both, instead of always running both like the paper's version.

LangGraph | LangChain | Neo4j | RAG | RPA | Python



Automated job recommendation system

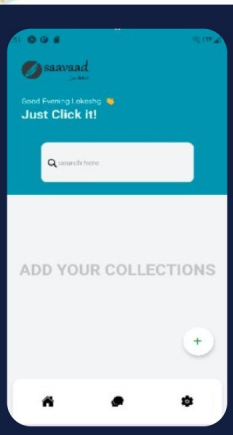
SSSIHL. Nov 2023 – Apr 2024

- Automated LinkedIn job searches with Selenium and RPA.
- Pulled required skills out of job descriptions with Sentence Transformers, which improved profile matching accuracy by 25%.
- Searched 500+ listings on Weaviate and emailed each user a personalised report.

Selenium | RPA | Sentence Transformers | Weaviate

PROJECTS, CONTINUED





SAAVAAD: summarisation and translation app

SSSIHL. Nov 2022 – Apr 2023

- Built a React Native app (Android and iOS) that summarises text and translates it into 10+ languages.
- Served the models through a Flask API, made public with Serveo.net.
- Added encrypted MongoDB storage, offline use and user accounts.

React Native | Flask | MongoDB

github.com/SAAVAAD

SMALLER PROJECTS

RAS Bot

Scores a startup's investment risk from a single form or a bulk CSV upload. Built at Sentinel AI ML Labs.

View on GitHub

LVAD candidacy predictor

Predicts whether a heart failure patient is a candidate for an LVAD, with SHAP explanations for each prediction.

View on GitHub

HeartCam AI

Real-time, video-based system for spotting early signs of cardiovascular distress.

View on GitHub

EDUCATION

MBA, Strategy and Consulting
Expected 2028

BITS Pilani, Work Integrated Learning Programme (for working professionals)

MSc, Data Science and Computing
Jul 2023 – May 2025

Sri Sathya Sai Institute of Higher Learning (SSSIHL)

CGPA 8.3. Thesis: hybrid question answering with LangGraph, multi-hop RAG and knowledge graphs.

Coursework: stochastic processes, big data analysis, deep learning, data engineering, linear algebra, optimisation, multivariate statistics, data visualisation, cloud computing, NLP.

BSc (Hons), Computer Science
Jun 2020 – Jun 2023

Sri Sathya Sai Institute of Higher Learning (SSSIHL)

CGPA 7.1

Coursework: data structures, software engineering, inferential statistics, high-performance computing, object-oriented programming, networking, discrete mathematics, web development, graph theory, DBMS.

Higher secondary, Mathematics and Computer Science
Jun 2018 – Mar 2020

PSG Sarvajana High School, Coimbatore

CERTIFICATIONS

- Neo4j Certified Professional
- Supervised Machine Learning: Regression and Classification
- Power BI
- Robotic Process Automation Student Essentials
- Diploma in Sathya Sai Balvikas
- AWS Academy Graduate: Cloud Foundations
- Databases and SQL for Data Science
- Software Engineer Intern certification
- Soft Skills Development (NPTEL)

SKILLS

AI and ML	Machine learning, deep learning, NLP, predictive modelling, speech recognition (ASR), explainable AI (SHAP), computer vision
LLMs and generative AI	RAG, agentic AI (LangGraph), LangChain, knowledge graphs, ontology design, hybrid vector and graph retrieval, LLM fine-tuning (Unsloth), text to Cypher, LLM evaluation (BLEU, ROUGE, faithfulness), small task-specific models
Programming	Python, SQL, R, JavaScript, TypeScript, C/C++, Java
Data and backend	pandas, NumPy, scikit-learn, PyTorch, TensorFlow, PySpark, Neo4j (Cypher), Weaviate, MongoDB, MySQL, FastAPI, Flask, AWS
Product and web	React, React Native, Node.js, Next.js, Android SDK, WordPress, HTML and CSS, Git
Strategy and operations	Financial modelling (capex, opex, P&L, IRR), market entry strategy, CDSCO and MDR 2017, ISO 13485, ABDM and ABHA, release management, pitch decks, Power BI, advanced Excel
Automation and media	RPA, Selenium, WhatsApp workflow automation, DaVinci Resolve, Premiere Pro, Photoshop, Canva

BEYOND WORK



Investiture ceremony, SSSIHL



Video editing and studio work



Annual sports meet, SSSIHL

CONFERENCES

Presented and pitched

Global Innovation Summit and Retreat

Colombo, Sri Lanka. 2–4 Sep 2025

Pitched and helped present a cardiac MedTech client's artificial heart project, for Sentinel AI ML Labs.

Presented

ICE on I: PHILIPPiNEXT, 1st International Conference and Exposition on Innovations

Okada Manila, Parañaque City, Philippines. 14–17 Jul 2025

Helped present a cardiac MedTech client's artificial heart project, for Sentinel AI ML Labs.

Attended

Sangam 2025 Pitch Fest, IIT Madras Alumni Summit

Taj MG Road, Bengaluru. 5 Jul 2025

Presented paper

13th International Conference on Contemporary Engineering and Technology (ICCET 2025)

Chennai, with Samarkand State University. 22–23 Mar 2025

SAGE: hybrid knowledge retrieval for Ayushman Bharat treatment guidelines.

HONOURS

Ranked 15th nationally in Analytics Olympiad 2023, a machine learning hackathon hosted by MachineHack	2023
Best Intern, Sentinel AI ML Labs	2024
Cleared the Actuarial Common Entrance Test (ACET)	Mar 2023
House Championship Award, SSSIHL	